

Lecture 7 Worksheet

Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

Lines and Planes in $\mathbb{R}^3$

1. A Line Through the Origin in Parametric Form

Let $\ell = \text{span} \left(\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

(a) Write ℓ in vector-parametric form.

$$\ell = \{ \quad \quad \quad \}.$$

(b) Write the corresponding scalar-parametric equations for x , y , and z .

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}},$$

$$z = \underline{\hspace{2cm}}.$$

(c) **Homework:** Graph ℓ using the Desmos 3D Calculator.

2. A Plane Through the Origin in Parametric Form

Let $P = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

(a) Write P explicitly in vector-parametric form using two parameters s and t .

$$P = \{ \quad \quad \quad \}.$$

(b) Write the corresponding scalar-parametric equations for x , y , and z .

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}},$$

$$z = \underline{\hspace{2cm}}.$$

(c) **Homework:** Graph P using the Desmos 3D Calculator.

3. An Equation for a Plane Through the Origin

Consider the plane from Question 2: $P = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

- (a) Find a nonzero vector $\vec{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ that is perpendicular to both spanning vectors.

Thus, $\vec{n}$ must satisfy $\vec{n} \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = 0$ and $\vec{n} \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 0$.

$$\vec{n} = \begin{bmatrix} \\ \\ \end{bmatrix}.$$

- (b) Use $\vec{n}$ to write an equation for P in dot-product form.

$$\vec{n} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underline{\hspace{2cm}}.$$

Then write the equation explicitly as a linear equation in x , y , and z .

- (c) **Homework:** Graph your equation for P using the Desmos 3D Calculator. Compare the graph with the one you made in Question 2(c).

4. A Line Through the Origin as the Solution of a Linear System

Consider the line from Question 1: $\ell = \text{span} \left(\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$. A single linear equation in $\mathbb{R}^3$ usually describes a plane, not a line. To describe ℓ using equations, we need two planes whose intersection is ℓ .

- (a) Find two vectors $\vec{n}_1$ and $\vec{n}_2$ that are both perpendicular to $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, but are not scalar multiples of each other.

$$\vec{n}_1 = \begin{bmatrix} \\ \\ \end{bmatrix}, \quad \vec{n}_2 = \begin{bmatrix} \\ \\ \end{bmatrix}.$$

Check your choices:

$$\vec{n}_1 \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \underline{\hspace{2cm}}, \quad \vec{n}_2 \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \underline{\hspace{2cm}}.$$

- (b) Use these vectors to write a system of two homogeneous linear equations whose solution set is ℓ .

$$\left\{ \begin{array}{l} \end{array} \right.$$

- (c) **Homework:** Graph both equations in your system using the Desmos 3D Calculator. The two planes should intersect in the line ℓ .

5. An Affine Line in Parametric Form

Let $\ell' = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

(a) Write ℓ' explicitly in vector-parametric form.

$$\ell' = \{ \quad \quad \quad \}.$$

(b) Write the corresponding scalar-parametric equations for x , y , and z .

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}},$$

$$z = \underline{\hspace{2cm}}.$$

(c) **Homework:** Graph ℓ' using the Desmos 3D Calculator.

6. An Affine Plane in Parametric Form

Let $P' = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

(a) Write P' explicitly in vector-parametric form using two parameters s and t .

$$P' = \{ \quad \quad \quad \}.$$

(b) Write the corresponding scalar-parametric equations for x , y , and z .

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}},$$

$$z = \underline{\hspace{2cm}}.$$

(c) **Homework:** Graph P' using the Desmos 3D Calculator.

7. An Equation for an Affine Plane

Consider the affine plane from Question 6: $P' = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$. Use the normal vector $\vec{n}$ that you found in Question 3.

A point $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ lies in P' when $\vec{n} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n} \cdot \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

(a) Compute the constant on the right-hand side.

$$\vec{n} \cdot \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = \underline{\hspace{2cm}}.$$

(b) Write an equation for P' in dot-product form.

$$\vec{n} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underline{\hspace{2cm}}.$$

Then write the equation explicitly as a linear equation in x , y , and z .

(c) **Homework:** Graph your equation for P' using the Desmos 3D Calculator. Compare the graph with the one you made in Question 6(c).

8. An Affine Line as the Solution of a Linear System

Consider the affine line from Question 5: $\ell' = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$. Use the two normal vectors $\vec{n}_1$ and $\vec{n}_2$ that you found in Question 4.

A point $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ lies on ℓ' when $\vec{n}_1 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n}_1 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $\vec{n}_2 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n}_2 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

(a) Compute the two constants on the right-hand sides.

$$\vec{n}_1 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \underline{\hspace{2cm}}, \quad \vec{n}_2 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \underline{\hspace{2cm}}.$$

(b) Write a system of two linear equations whose solution set is ℓ' .

$$\left\{ \begin{array}{l} \\ \end{array} \right.$$

(c) **Homework:** Graph both equations in your system using the Desmos 3D Calculator. The two planes should intersect in the affine line ℓ' . Compare the intersection with the graph from Question 5(c).